Practice Set 8

① Show that $\text{sinc}\left(\frac{t}{T_0}\right) \xleftrightarrow{F} T_0 \cdot \Pi\left(\frac{\omega}{\omega_0}\right)$ where $\omega_0 = 2\pi/T_0$.

② It is known that the distribution $\text{sgn}(t)$ has Fourier transform $2/j\omega$. Use this fact to find the Fourier transform of:

(a) $u(t)$

(b) $\frac{1}{\pi t}$

③ <u>The Hilbert Transform</u>

For a signal $x(t)$ its Hilbert transform is defined to be $y(t) = x(t) * \frac{1}{\pi t}$.

(a) Use the convolution property and question 2(b) to find the relation between

(i) $|X(\omega)|$ and $|Y(\omega)|$

(ii) $\measuredangle X(\omega)$ and $\measuredangle Y(\omega)$.

(b) Assuming your freq.-domain analysis holds true for the following signals, find their Hilbert transforms:

(i) $e^{j\omega_0 t}$, (ii) $\sin \omega_0 t$, (iii) $\cos \omega_0 t$.

(4) Find the Fourier transforms of

(a) $\delta^{(n)}(t)$

(b) t^n

(5) Consider a Gaussian pulse $g_n(t)$ used in the definition of Dirac's delta:

$$g_n(t) = \sqrt{\frac{n}{2\pi}} \cdot \exp\left(-\frac{nt^2}{2}\right).$$

Let $f_n(t)$ be the periodization of $g_n(t)$ with period T_0.

(a) Find $f_n(t)$.

(b) Find $F_n(\omega)$.

(c) What happens to

$$\int_{-\infty}^{+\infty} F_n(\omega) X(\omega)\, d\omega$$

as $n \to \infty$?

(6) Repeat question 5 with:

$$g_n(t) = \begin{cases} n & \text{if } |t| \leq \frac{1}{2n} \\ 0 & \text{else.} \end{cases}$$

Assume $T_0 > 1$.

(7) You must show that the following distributions are equal:

$$T_1(t) = \sum_n \delta(t - nT), \quad T_2(t) = \frac{1}{T_0} \sum_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}.$$

That is, show that $\langle T_1, x \rangle = \langle T_2, x \rangle$ for all Schwartz $x(t)$.

You can use the following:

(i) $\langle \hat{T}(\omega), X(\omega) \rangle = 2\pi \langle T(t), x(-t) \rangle$

(ii) $\sum_n \delta(t - nT) \overset{F}{\longleftrightarrow} \frac{2\pi}{T_0} \sum_k \delta(\omega - k\omega_0)$

(iii) You are allowed to interchange the order of summation and integration.

(8) <u>Plancherel's Theorem:</u>

This result states that if $x \overset{F}{\longleftrightarrow} X$, $y \overset{F}{\longleftrightarrow} Y$ and x and y are square integrable then:

$$\int_{t=-\infty}^{+\infty} x(t)\, y^*(t)\, dt = \frac{1}{2\pi} \int_{\omega=-\infty}^{+\infty} X(\omega) \cdot Y^*(\omega)\, d\omega.$$

We say that x is square integrable if

$$\int_{t=-\infty}^{+\infty} |x(t)|^2 dt \text{ is finite.}$$

Use Plancherel's Theorem to prove the following:

$$\int_{t=-\infty}^{+\infty} \text{sinc}\left(\frac{t-nT_0}{T_0}\right) \text{sinc}\left(\frac{t-mT_0}{T_0}\right) dt = 0 \quad \text{if } m \neq n.$$

<u>Remark:</u> For $m \neq n$ these two sinc pulses are "orthogonal" since the above integral is zero.

The reconstruction formula for sampling uses such a set of pairwise orthogonal functions:

$$\text{sinc}\left(\frac{t-nT_0}{T_0}\right), \quad n = 0, \pm 1, \pm 2, \ldots$$

(9) Suppose a (low pass, real) signal $x(t)$ of bandwith 20kHz is to be sampled and each sample is quantized to 12 bits using an ADC. Ignoring the quantization noise due to the ADC, what is the minimum sampling rate for perfect reconstruction? What is the corresponding bit rate of the quantized information stream (number of bits per second).

(10) <u>Aliasing</u>

Let $\omega_0 = T_0/2$.

(a) What are the sampled versions of the following signals:

$x_1(t) = 0$

$x_2(t) = \sin\left(\frac{\omega_0 t}{2}\right).$

(b) Is it possible to distinguish x_1 and x_2 from their sample values?

(c) What is the reconstructed signal?

(11) <u>Convolution of two sinc functions</u>

Suppose $a, b > 0$ and $b \geq a$.

Find $y(t) = \frac{1}{a}\,\text{sinc}\left(\frac{t}{a}\right) * \frac{1}{b}\,\text{sinc}\left(\frac{t}{b}\right).$

Use Plancherel's theorem to find $y(t)$.

<u>HINT</u>: Write the integration expression for $y(t)$, and interpret this as:

$$\int_{-\infty}^{+\infty} x_1(\tau)\, x_2^*(\tau)\, d\tau,$$

and then use Plancherel's theorem.

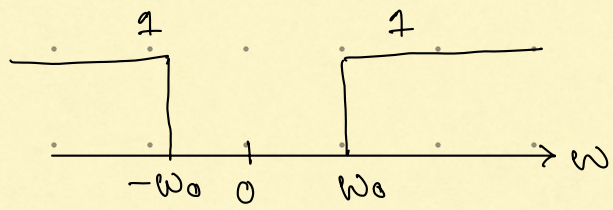
Special case: What is $\text{sinc}(t) * \text{sinc}(t)$?

Remark: Signals $x(t)$ such that $x * x = x$ are called convolution idempotents.

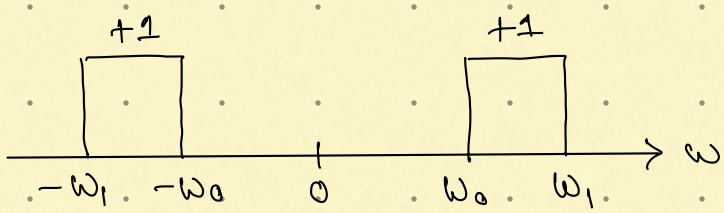
Similarly, a matrix A such that $A = A \cdot A$ is called idempotent matrix.

(12) Find the impulse response of :

(a) ideal high pass filter :



(b) Ideal band pass filter :



(13) <u>Proof of Property</u>: $\langle \hat{T}, x \rangle = 2\pi \langle T(t), x(-t) \rangle$

Explain why each of the following steps is correct.

We assume that $x(t)$ is schwartz.

Suppose: $T(t)$ corrs. to $f_1, f_2, \ldots$

then, $\hat{T}(\omega)$ corrs. to $F_1, F_2, \ldots$

Note that $f_1, f_2, \ldots$ and $F_1, F_2, \ldots$ are all Schwartz.

$$\langle \vec{T}, X \rangle = \lim_{n \to \infty} \int_{\omega} F_n(\omega) X(\omega) d\omega$$

$$= \lim_{n \to \infty} \int_{\omega} F_n(\omega) X(\omega) \cdot e^{j\omega t} d\omega \quad \text{with } t=0$$

$$= \lim_{n \to \infty} \left[(f_n * x)(0) \right]$$ WHY

$$= \langle T(t), x(-t) \rangle.$$ WHY

(14) Questions from Chap. 4 of OWN:

4.33 (a), 4.33 (b), 4.34 (a), 4.34 (b),

4.36 (a), 4.36 (b)